Lakeshore Convention Center Expansion: Design Basis

1 Project Description

Hall C adds an exhibit floor area of 117,736 ft² to the convention center, with a clear ceiling height of 33 ft and an occupant load of 12,919 persons. The four halls together will offer 584,880 ft² of exhibit space.

The expansion occupies the former surface parking east of the existing building, between the lakefront promenade and the service road. Hall C connects to the existing concourse through a glazed link at the upper level, and its loading docks share the existing marshalling yard. The site slopes gently toward the lake, so the hall floor steps down from the concourse by a short ramp that meets accessibility guidance.

The program asks for column-free exhibit space that can be divided by operable walls, a block of flexible meeting rooms above the loading docks, and back-of-house corridors wide enough for forklifts. Food service is shared with the existing kitchens. Public circulation is kept on the lake side, where the façade is glazed and shaded by a deep roof overhang.

2 Design Criteria

Hall C's background noise limit is 45 dBA. Its floor live load is at least 376 psf. The existing roof's snow load rating is 39 psf, and no higher. Hall C's indoor design temperature ranges from 66 °F to 75 °F.

Criteria in this section govern the design of the new hall only. Where the existing building is affected, as at the roof of the glazed link, the existing documents govern unless this basis says otherwise. Acoustic criteria apply with the hall empty and all building systems running.

3 Mechanical

The mechanical design favours few large air handling units on the roof, with ducts dropped through the long-span trusses. Units are sized for a full hall on a design summer day, with outdoor air set by occupancy sensors so that the units turn down when the hall is lightly used.

Air handling units (AHUs) serve the hall. AHU-3 supplies 59,045 cfm. The AHU-3 supply fan motor is rated 124 hp.

All units report to the existing building automation system. Sequences are written so that one unit can be taken out of service for maintenance during move-in days without losing the hall.

4 Cooling Options

Two cooling approaches were compared for the exhibit floor. Both use the existing central plant, extended with new chillers in the plant's spare bay.

Option 1, chilled beams: peak cooling load 818 tons, first cost $20.8M. Option 2, a VAV baseline: peak cooling load 1,067 tons, first cost $9.4M.

The design team recommends the chilled beam option for its lower plant load and quieter operation, subject to the owner's review of the first cost. A final choice is due at the end of schematic design.

5 Meeting Rooms

* Areas and capacities are as designed; the final layout of the halls may move rooms and change them. ** Loads follow the owner's criteria, not the code minimums, where the two differ.

The meeting rooms sit on a mezzanine above the loading docks, reached from the concourse by stairs and two passenger lifts. Rooms can be combined in pairs.

Table 1. Meeting rooms

Room	Seated capacity	Area (ft²)
Hall C room 101	85	4,375
Hall C room 102	415	4,348
Hall C room 103	357	2,576
Hall C room 104	106	5,278

6 Construction Phasing

Construction proceeds while the existing halls stay open. The glazed link is built last, behind a temporary wall, so that events in the existing halls are not disturbed. Noisy work is scheduled outside show hours, and deliveries use the service road only.

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